

# Multivariable Calculus

MATH 1920 Lecture Notes

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# 1 Vectors in Two and Three Dimensions

**Question:** What information specifies a line?

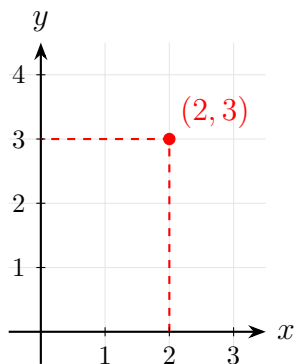
- 2 points
- A point and a direction vector
- Slope and  $y$ -intercept
- Intersection of 2 non-parallel planes

**Comparison:**

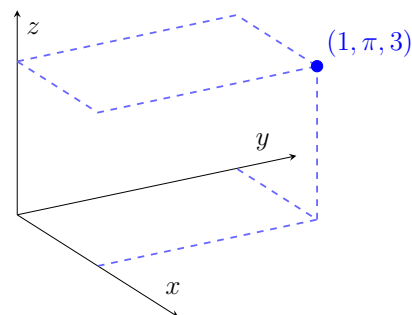
- **Single-variable calculus:**  $f(x) \rightarrow$  rate of change of  $x$
- **Multivariable calculus:**  $f(x, y) \rightarrow$  functions of multiple variables

## 1.1 Coordinates

**2-dimension (2D):**  $(2, 3)$



**3-dimension (3D):**  $(1, \pi, 3)$



## 1.2 Vectors and Position Vectors

Given points  $P = (1, 3)$ ,  $Q = (5, 4)$  and  $P' = (1, 1)$ ,  $Q' = (5, 2)$ :

### Definition 1.1: Equivalent Vectors

Two vectors are **equivalent** if one is a translation (pure shift without rotation) of the other.

**Example 1.1.** Let  $R = (4, 1)$  and  $O = (0, 0)$ . Then:

$$\overrightarrow{PQ} = \overrightarrow{P'Q'} = \overrightarrow{OR}$$

### Definition 1.2: Position Vector

A vector whose initial point is at the origin  $O(0, 0)$  is called a **position vector**. It is written using angle brackets:

$$\overrightarrow{OR} = \langle 4, 1 \rangle$$

### Definition 1.3: Magnitude / Norm

The **length** or **magnitude** of a vector  $\vec{v}$ , denoted by  $\|\vec{v}\|$ , is the length of the line segment between its endpoints.

If  $\vec{v} = \langle v_1, v_2 \rangle$ , then its magnitude is given by:

$$\|\vec{v}\| = \sqrt{v_1^2 + v_2^2}$$

For example, using  $\overrightarrow{PQ} = \langle 4, 1 \rangle$ :

$$\|\overrightarrow{PQ}\| = \sqrt{4^2 + 1^2} = \sqrt{17}$$

## 1.3 Unit Vectors

A vector  $\vec{v}$  is called a **unit vector** if its magnitude is 1, i.e.,  $\|\vec{v}\| = 1$ .

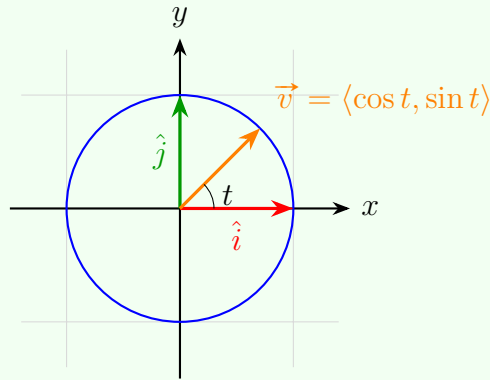
### Problem 1.1: Describing Unit Vectors

Describe all unit vectors in  $\mathbb{R}^2$ .

### Solution 1.1

A vector  $\vec{v} = \langle x, y \rangle$  is a unit vector if and only if  $x^2 + y^2 = 1$ . Equivalently, using polar parameterization:

$$\vec{v}(t) = \langle \cos t, \sin t \rangle \quad \text{for } 0 \leq t < 2\pi$$



**Standard Basis Vectors:**

- **2D Basis Vectors:**  $\hat{i} = \langle 1, 0 \rangle$ ,  $\hat{j} = \langle 0, 1 \rangle$
- **3D Basis Vectors:**  $\hat{i} = \langle 1, 0, 0 \rangle$ ,  $\hat{j} = \langle 0, 1, 0 \rangle$ ,  $\hat{k} = \langle 0, 0, 1 \rangle$

## 1.4 Vector Operations

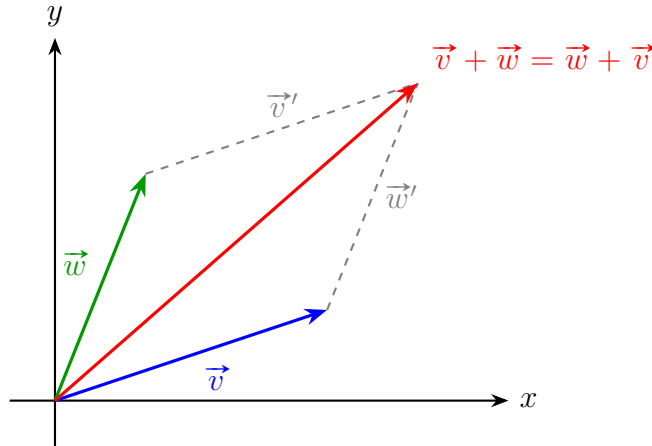
**Addition and Subtraction:**

$$\langle v_1, v_2 \rangle + \langle w_1, w_2 \rangle = \langle v_1 + w_1, v_2 + w_2 \rangle$$

$$\langle v_1, v_2 \rangle - \langle w_1, w_2 \rangle = \langle v_1 - w_1, v_2 - w_2 \rangle \quad (\text{Equivalent to } \vec{v} + (-1)\vec{w})$$

**Scalar Multiplication:** For any scalar  $\lambda \in \mathbb{R}$  and vector  $\vec{v} = \langle v_1, v_2 \rangle$ :

$$\lambda \vec{v} = \langle \lambda v_1, \lambda v_2 \rangle$$



### Important Note

Vectors are **not** designed to be divided! (Vector division is undefined).

## 1.5 Parallel Vectors

### Definition 1.4: Parallel Vectors

Two non-zero vectors  $\vec{v}$  and  $\vec{w}$  ( $\vec{v}, \vec{w} \neq \langle 0, 0 \rangle$ ) are *parallel* (denoted as  $\vec{v} \parallel \vec{w}$ ) if there exists a non-zero scalar  $\lambda \in \mathbb{R}$  such that:

$$\vec{w} = \lambda \vec{v}$$

**Equivalent Geometric Definition:**  $\vec{v} \parallel \vec{w}$  if and only if both vectors define or lie on the same line passing through the origin.

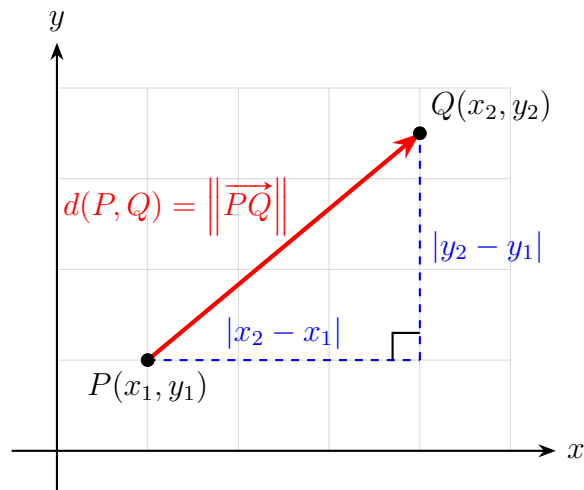
## 2 Discussion Week 1: Distance Formulae and Geometric Shapes

### Definition 2.1: Distance Formula in $\mathbb{R}^2$

Let  $(x, y) \in \mathbb{R}^2$  and points  $P = (x_1, y_1)$ ,  $Q = (x_2, y_2)$ . The distance between  $P$  and  $Q$  is given by:

$$d(P, Q) = \left\| \overrightarrow{PQ} \right\| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

This distance formula is a direct application of the Pythagorean theorem, where the distance  $d(P, Q)$  forms the hypotenuse of a right-angled triangle with legs  $|x_2 - x_1|$  and  $|y_2 - y_1|$ , as shown in the figure below.



The case in  $\mathbb{R}^3$  follows similarly:

### Definition 2.2: Distance Formula in $\mathbb{R}^3$

Let  $(x, y, z) \in \mathbb{R}^3$  and points  $P = (x_1, y_1, z_1)$ ,  $Q = (x_2, y_2, z_2)$ . The distance between  $P$  and  $Q$  is given by:

$$d(P, Q) = \|\vec{PQ}\| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

## 2.1 Equations of Geometric Shapes

Consider the equation  $x^2 + y^2 = 4$ .

In  $\mathbb{R}^2$ , this equation is equivalent to the set:

$$\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 4\}$$

This represents a circle with radius  $r = 2$  centered at the origin. The distance from the origin  $(0, 0)$  to any point  $(x, y)$  on this circle is constant and equal to 2, which can be verified

directly using the distance formula:

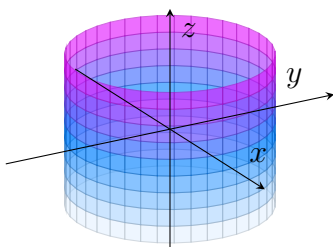
$$\sqrt{(x-0)^2 + (y-0)^2} = 2$$

Hence, the geometry of the circle is directly implied by the Pythagorean theorem.

Now consider the equation  $x^2 + y^2 = 1$  in three-dimensional space,  $\mathbb{R}^3$ . This set is formally expressed as:

$$\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = 1\}$$

Because  $z$  is completely unrestricted, the cross-section at any constant height  $z = c$  forms the unit circle  $x^2 + y^2 = 1$ . Consequently, this equation describes a *right circular cylinder* extending infinitely along the  $z$ -axis, as shown.



## 2.2 Linear Equations across Dimensions

In  $\mathbb{R}^2$ , the general form of a linear equation is given by:

$$Ax + By + C = 0 \quad (A, B, C \in \mathbb{R}, \text{ with } A \text{ and } B \text{ not both zero})$$

This equation represents a *line* in two-dimensional space.

In  $\mathbb{R}^3$ , however, the exact same algebraic equation  $Ax + By + C = 0$  represents a *plane* parallel to the  $z$ -axis (since  $z$  is a free variable), not a line.

To uniquely define a line in three-dimensional space, a single linear equation is insufficient; we require a system of *two* independent linear equations (representing the intersection of two non-parallel planes).



### 3 Vector and Directions

#### Problem 3.1: Vector Directions

Let  $\vec{v} = \langle 3, -4, 2 \rangle$ . Find a unit vector in the same direction as  $\vec{v}$ .

#### Solution 3.1

To find a unit vector in the same direction as  $\vec{v}$ , we first compute its magnitude:

$$\|\vec{v}\| = \sqrt{3^2 + (-4)^2 + 2^2} = \sqrt{9 + 16 + 4} = \sqrt{29}$$

The unit vector in the same direction as  $\vec{v}$  is then:

$$\hat{v} = \frac{\vec{v}}{\|\vec{v}\|} = \frac{1}{\sqrt{29}} \langle 3, -4, 2 \rangle$$

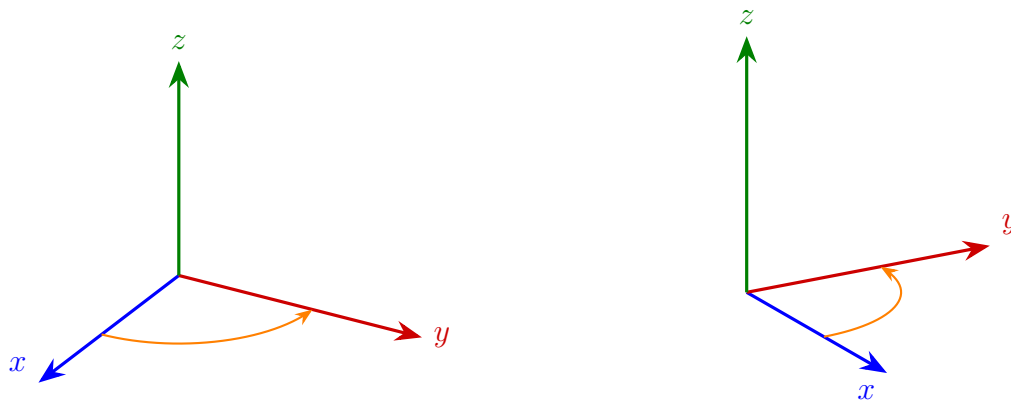
In general, for any non-zero vector  $\vec{w}$ ,

$$\begin{aligned}\|\lambda \vec{w}\| &= \sqrt{(\lambda w_1)^2 + (\lambda w_2)^2 + (\lambda w_3)^2} \\ &= \sqrt{\lambda^2(w_1^2 + w_2^2 + w_3^2)} \\ &= |\lambda| \|\vec{w}\|\end{aligned}$$

Hence, if we want a unit vector in the same direction as  $\vec{w}$ , we can choose  $\lambda = \frac{1}{\|\vec{w}\|}$ , giving us:

$$\hat{w} = \frac{\vec{w}}{\|\vec{w}\|}$$

**Right-Hand Rule** The relative position of the  $x$ -,  $y$ -, and  $z$ -axes in  $\mathbb{R}^3$  is determined by the *right-hand rule*. If you point your right thumb along the positive  $x$ -axis and your index finger along the positive  $y$ -axis, then your middle finger (perpendicular to both) will point along the positive  $z$ -axis.



### Definition 3.1: Parameterizations

A parameterization is a function whose range is a given geometric object.

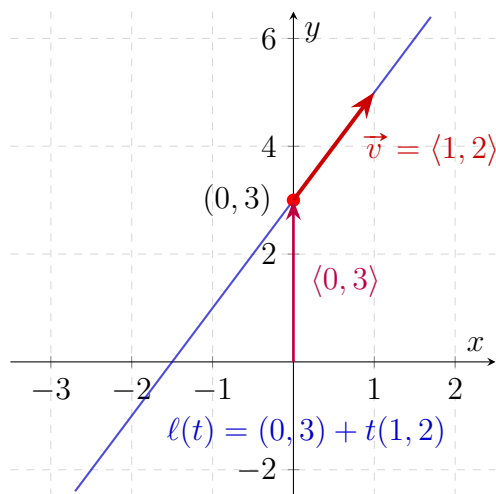
For example, a parameterization of the unit circle is given by:

$$f(t) = (\cos t, \sin t)$$

**Example 3.1.** To parameterize the line  $y = 2x + 3$ :

$$\begin{aligned}\ell(t) &= (t, 2t + 3) \\ &= \langle 0, 3 \rangle + \langle t, 2t \rangle\end{aligned}$$

The plot is then given:



To generalize, to parameterize a line in  $\mathbb{R}^2$ , we can first take a point on the line. Then, we

take some  $\vec{v}$  in the direction of the line. Then, we can write the parameterization as:

$$\ell(t) = (p_1 + v_1t, p_2 + v_2t)$$

In  $\mathbb{R}^3$ , we can do the same thing.

**Example 3.2.** Parameterize the line passing through  $P(1, 2, 0)$  and  $Q(5, -1, 1)$ .

$$\begin{aligned}\overrightarrow{PQ} &= \langle 5 - 1, -1 - 2, 1 - 0 \rangle \\ &= \langle 4, -3, 1 \rangle \\ \ell(t) &= (1, 2, 0) + t(4, -3, 1) \\ &= (1 + 4t, 2 - 3t, t)\end{aligned}$$

### 3.1 Dot Product

#### Definition 3.2: Dot Product

The dot product of two vectors  $\vec{v} = \langle v_1, v_2 \rangle$  and  $\vec{w} = \langle w_1, w_2 \rangle$  is defined as:

$$\vec{v} \cdot \vec{w} = v_1w_1 + v_2w_2$$

**Example 3.3.**

$$\begin{aligned}\vec{v} \cdot \vec{v} &= v_1^2 + v_2^2 = \|\vec{v}\|^2 \\ \vec{i} \cdot \vec{j} &= 1 \cdot 0 + 0 \cdot 1 = 0\end{aligned}$$

#### Problem 3.2: Dot Product and Angle

Find pairs of vectors with:

- Dot product  $> 0$
- Dot product  $= 0$
- Dot product  $< 0$

### Solution 3.2

- $\vec{v} \cdot \vec{v} = \|\vec{v}\|^2 > 0$  for any non-zero vector  $\vec{v}$ .
- $\vec{i} \cdot \vec{j} = 0$  since they are orthogonal.
- $\langle -1, -1 \rangle \cdot \langle 1, 1 \rangle = -1 - 1 = -2 < 0$ .

To conclude, the dot products of any two vectors corresponds to the following situations:

- Dot product = 0 if and only if the two vectors are orthogonal (perpendicular).
- Dot product  $> 0$  if the angle between the two vectors is acute (less than 90 degrees).
- Dot product  $< 0$  if the angle between the two vectors is obtuse (greater than 90 degrees).

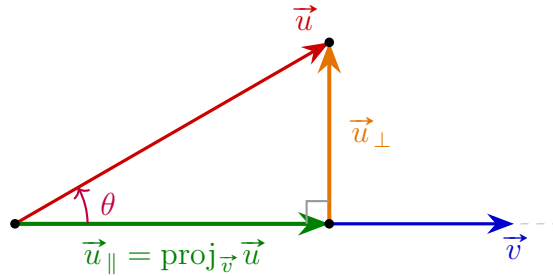
This is because  $\vec{v} \cdot \vec{w} = \|\vec{v}\| \|\vec{w}\| \cos \theta$ , where  $\theta$  is the angle between the vectors. Hence,

$$\theta = \arccos \left( \frac{\vec{v} \cdot \vec{w}}{\|\vec{v}\| \|\vec{w}\|} \right)$$

### Definition 3.3: Orthogonal Projection

Write  $\vec{u} = \vec{u}_{\parallel} + \vec{u}_{\perp}$ , where  $\vec{u}_{\parallel} \parallel \vec{v}$  and  $\vec{u}_{\perp} \perp \vec{u}_{\parallel}$ . Then,  $\vec{u}_{\parallel}$  is called the **orthogonal projection** of  $\vec{u}$  onto  $\vec{v}$ .

**Example 3.4.** Below is an illustration of orthogonal projection:



By this, we could see that

$$\vec{u} = \vec{u}_{\parallel} + \vec{u}_{\perp}$$

Then, we solve for  $\vec{u}_{\parallel}$  and  $\vec{u}_{\perp}$ .

Taking the dot product with  $\vec{v}$ :

$$\begin{aligned}
 \vec{u} \cdot \vec{v} &= (\vec{u}_{\parallel} + \vec{u}_{\perp}) \cdot \vec{v} \\
 &= \vec{u}_{\parallel} \cdot \vec{v} + \vec{u}_{\perp} \cdot \vec{v} \\
 &= \vec{u}_{\parallel} \cdot \vec{v} + 0 \quad (\text{since } \vec{u}_{\perp} \perp \vec{v}) \\
 &= \vec{u}_{\parallel} \cdot \vec{v}
 \end{aligned}$$

Let  $\vec{e}_v = \frac{1}{\|\vec{v}\|} \vec{v}$  be the unit vector in the direction of  $\vec{v}$ , and write  $\vec{u}_{\parallel} = \lambda \vec{e}_v$ . Then:

$$\begin{aligned}
 \vec{u} \cdot \vec{v} &= (\lambda \vec{e}_v) \cdot (\|\vec{v}\| \vec{e}_v) \\
 &= \lambda \|\vec{v}\| (\vec{e}_v \cdot \vec{e}_v) \\
 &= \lambda \|\vec{v}\| \|\vec{e}_v\|^2 \\
 &= \lambda \|\vec{v}\| \quad (\text{since } \|\vec{e}_v\| = 1)
 \end{aligned}$$

Solving for  $\lambda$ :

$$\lambda = \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|}$$

Thus, the parallel projection vector  $\vec{u}_{\parallel}$  and the orthogonal component  $\vec{u}_{\perp}$  are given by:

$$\begin{aligned}
 \vec{u}_{\parallel} &= \left( \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|} \right) \vec{e}_v = \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|^2} \vec{v} = \frac{\vec{u} \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \vec{v} \\
 \vec{u}_{\perp} &= \vec{u} - \vec{u}_{\parallel} = \vec{u} - \frac{\vec{u} \cdot \vec{v}}{\vec{v} \cdot \vec{v}} \vec{v}
 \end{aligned}$$

Furthermore, the magnitude of the parallel projection is:

$$\|\vec{u}_{\parallel}\| = \|\vec{u}\| |\cos \theta|$$

### Problem 3.3: Vector Projection

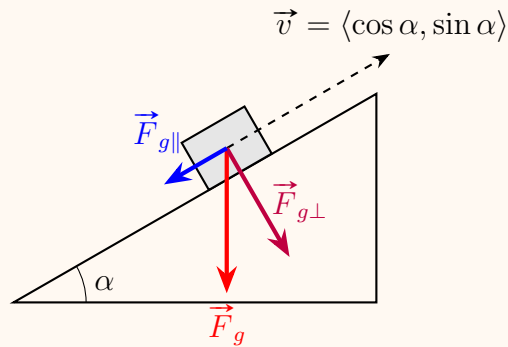
Find the projection of  $\vec{u} = \langle 3, 2, 2 \rangle$  onto  $\vec{v} = \langle 1, -2, 0 \rangle$

### Solution 3.3

$$\begin{aligned}\vec{u}_{\parallel} &= \frac{3 \cdot 1 + 2 \cdot (-1) + 2 \cdot 0}{1^2 + (-1)^2 + 0^2} \langle 1, -1, 0 \rangle \\ &= \frac{1}{2} \langle 1, -1, 0 \rangle \\ &= \left\langle \frac{1}{2}, -\frac{1}{2}, 0 \right\rangle\end{aligned}$$

### Problem 3.4: Acceleration on an Inclined Plane

Consider a block sliding down an inclined plane with an angle of inclination  $\alpha$ .



### Solution 3.4

To find the magnitude of acceleration down the incline, we project the gravitational force  $\vec{F}_g$  along the incline direction:

$$\begin{aligned}\text{acceleration} &= \left\| \vec{F}_{g\parallel} \right\| \\ &= \left\| \vec{F}_g \right\| |\cos \theta| \\ &= \left\| \vec{F}_g \right\| \sin \alpha = 9.8 \sin \alpha \quad (\text{m/s}^2)\end{aligned}$$

### Problem 3.5: Vector Projection

Find the projection of  $\vec{u} = \langle 3, 2, 2 \rangle$  onto  $\vec{v} = \langle 1, -1, 0 \rangle$ .

### Solution 3.5

Using the vector projection formula:

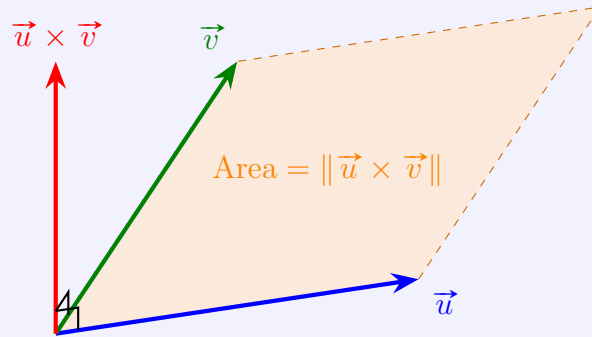
$$\begin{aligned}\vec{u}_{\parallel} &= \frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|^2} \vec{v} \\ &= \frac{3(1) + 2(-1) + 2(0)}{1^2 + (-1)^2 + 0^2} \langle 1, -1, 0 \rangle \\ &= \frac{3 - 2 + 0}{1 + 1 + 0} \langle 1, -1, 0 \rangle \\ &= \frac{1}{2} \langle 1, -1, 0 \rangle \\ &= \left\langle \frac{1}{2}, -\frac{1}{2}, 0 \right\rangle\end{aligned}$$

## 3.2 Cross Product

### Definition 3.4: Vector Cross Product

Let  $\vec{u}, \vec{v} \in \mathbb{R}^3$ . The **cross product**  $\vec{u} \times \vec{v}$  is a vector in  $\mathbb{R}^3$  satisfying the following three geometric properties:

1. **Orthogonality:**  $\vec{u} \times \vec{v} \perp \vec{u}$  and  $\vec{u} \times \vec{v} \perp \vec{v}$ .
2. **Magnitude:**  $\|\vec{u} \times \vec{v}\| = \|\vec{u}\| \|\vec{v}\| \sin \theta$ , which equals the area of the parallelogram formed by  $\vec{u}$  and  $\vec{v}$ .
3. **Orientation:** The ordered triple  $(\vec{u}, \vec{v}, \vec{u} \times \vec{v})$  is right-handed (positively oriented).



From Property (2), we have for the standard basis vectors:

$$\|\vec{i} \times \vec{i}\| = \|\vec{i}\| \|\vec{i}\| \sin 0^\circ = 0 \implies \vec{i} \times \vec{i} = \vec{0}$$

And by the Right-Hand Rule:

$$\vec{i} \times \vec{j} = \vec{k}$$